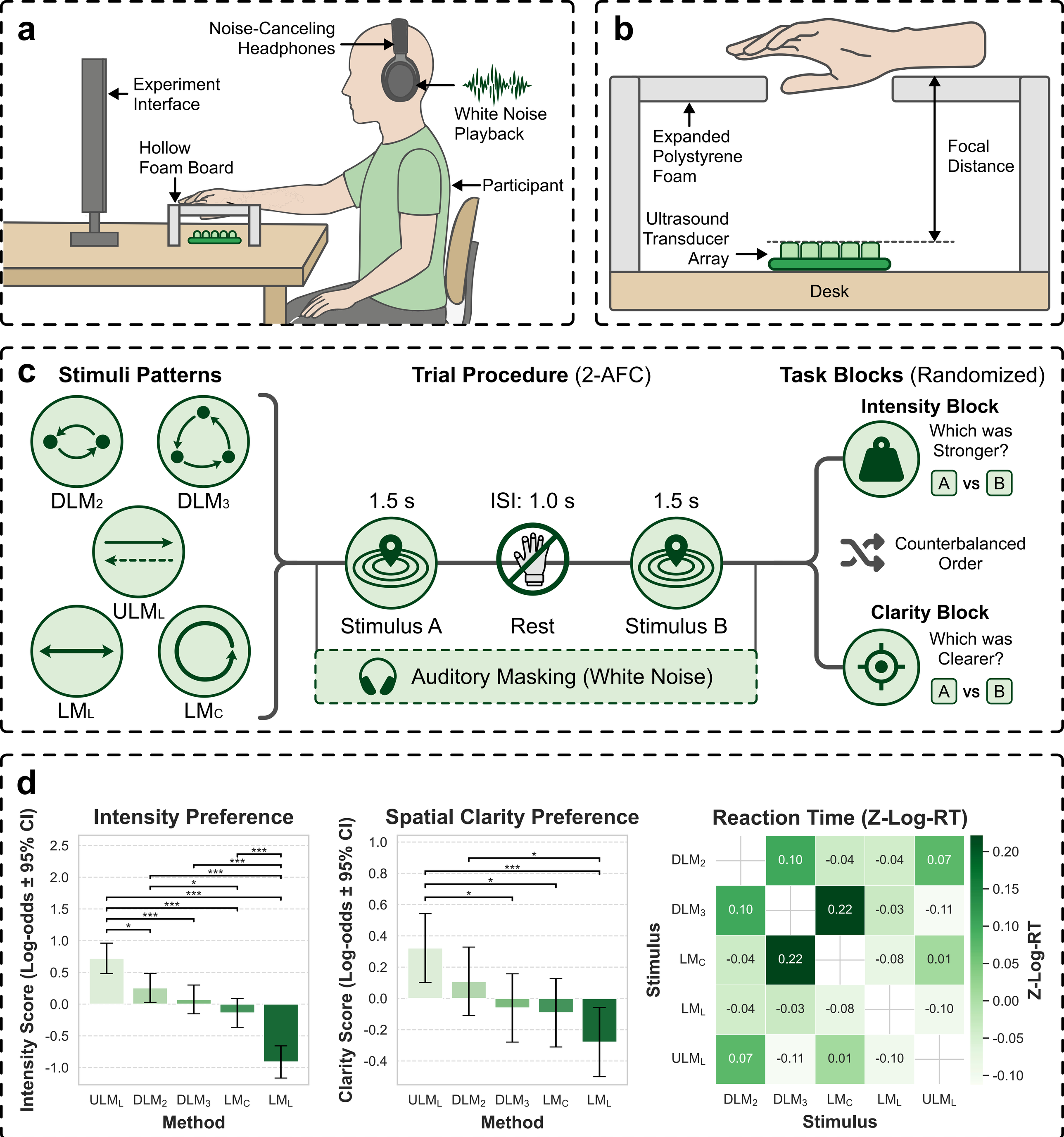
a
Noise-Canceling Headphones
Experiment Interface
White Noise Playback
Hollow Foam Board
Participant
b
Expanded Polystyrene Foam
Focal Distance
Ultrasound Transducer Array
Desk
c
Stimuli Patterns
DLM2
DLM3
ULML
LML
LMC
Trial Procedure (2-AFC)
1.5 s
ISI: 1.0 s
1.5 s
Stimulus A
Rest
Stimulus B
Auditory Masking (White Noise)
Task Blocks (Randomized)
Intensity Block
Which was Stronger?
A vs B
Counterbalanced Order
Clarity Block
Which was Clearer?
A vs B
d
Intensity Preference
Intensity Score (Log-odds ± 95% CI)
Method
Spatial Clarity Preference
Clarity Score (Log-odds ± 95% CI)
Method
Reaction Time (Z-Log-RT)
Stimulus
Z-Log-RT
DLM2 0.10 -0.04 -0.04 0.07
DLM3 0.10 0.22 -0.03 -0.11
LMC -0.04 0.22 -0.08 0.01
LML -0.04 -0.03 -0.08 -0.10
ULML 0.07 -0.11 0.01 -0.10